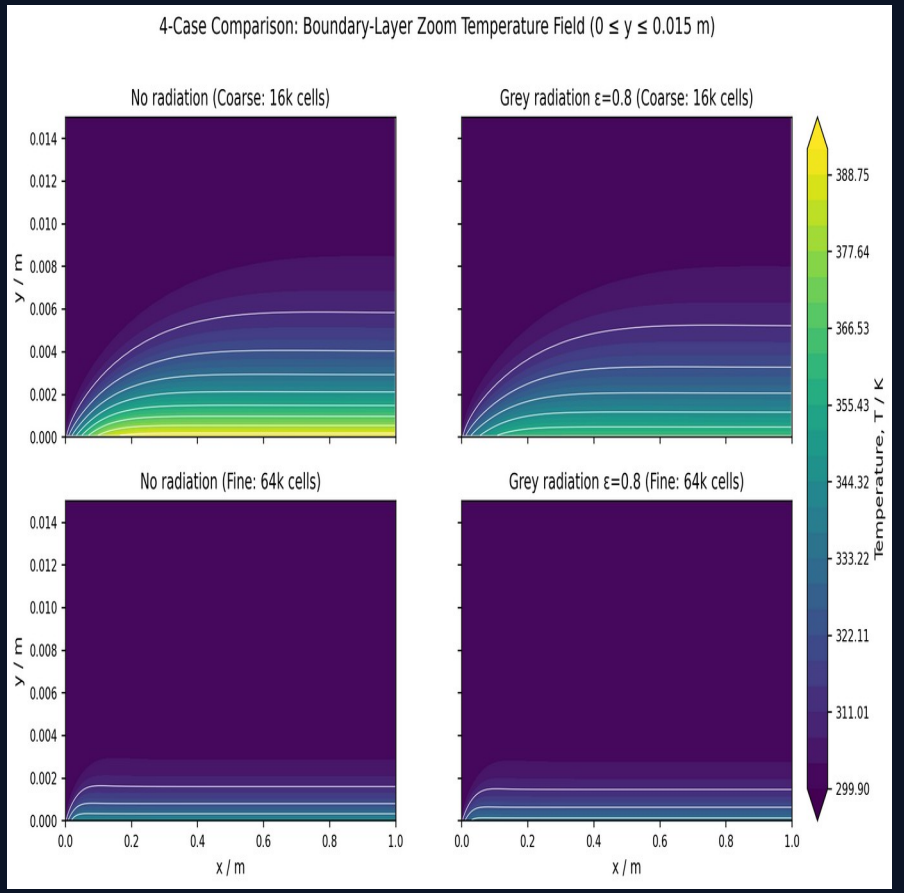
BOUNDARY LAYER CFD | OPENFOAM 13

Surface Radiation Cuts Flat-Plate Wall Temperature

Coupled convection-radiation CFD, validated against Blasius similarity



[Presenter name] | [Affiliation] | [Event / date]

One headline, four numbers

Re_L

1.26e5

Laminar, zero pressure gradient

64,000

hex cells

Fine grid, 80:1 wall grading

ε = 0.8

grey diffuse

Plate radiates to 300 K enclosure

[value]

[unit]

[caption: finalise after grid study]

Takeaway sentence goes here: one claim, one number, one reason to care.

Figure left, claim right

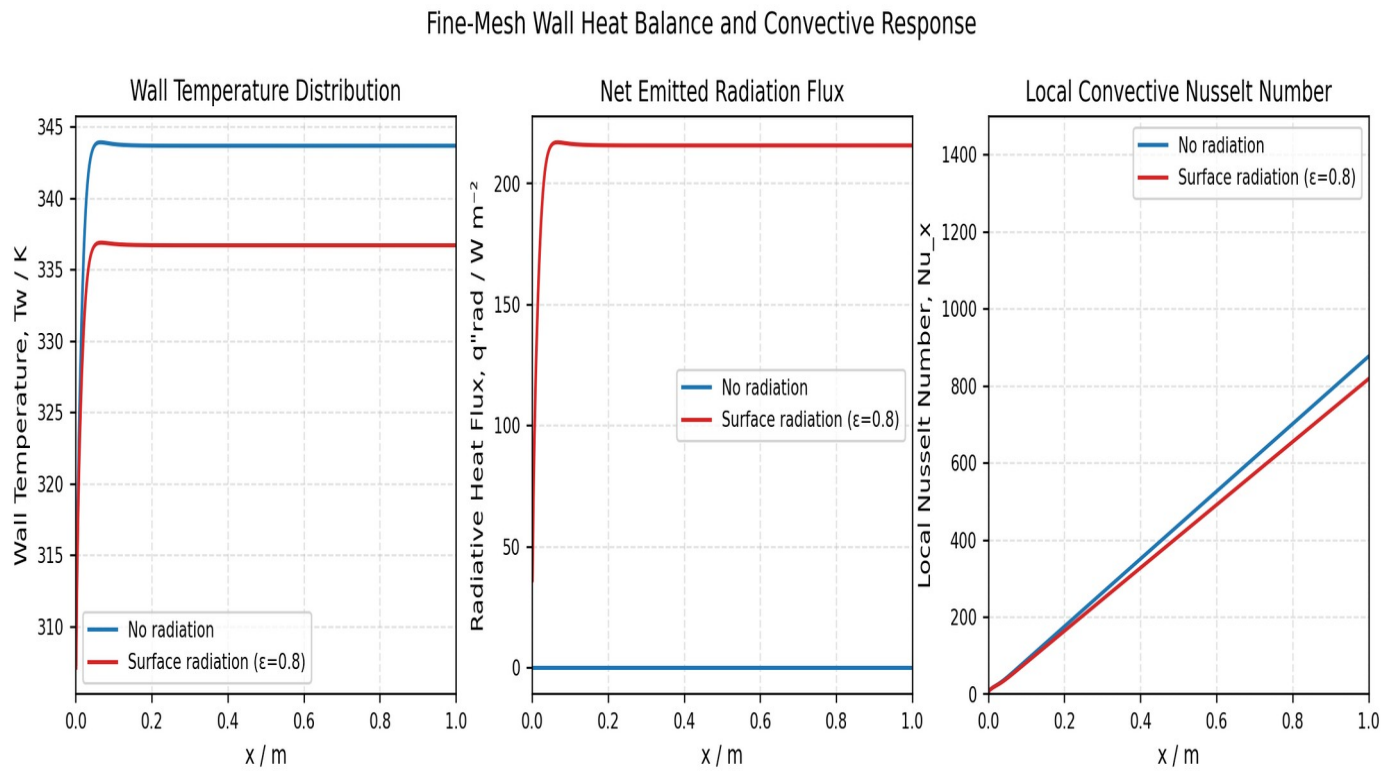


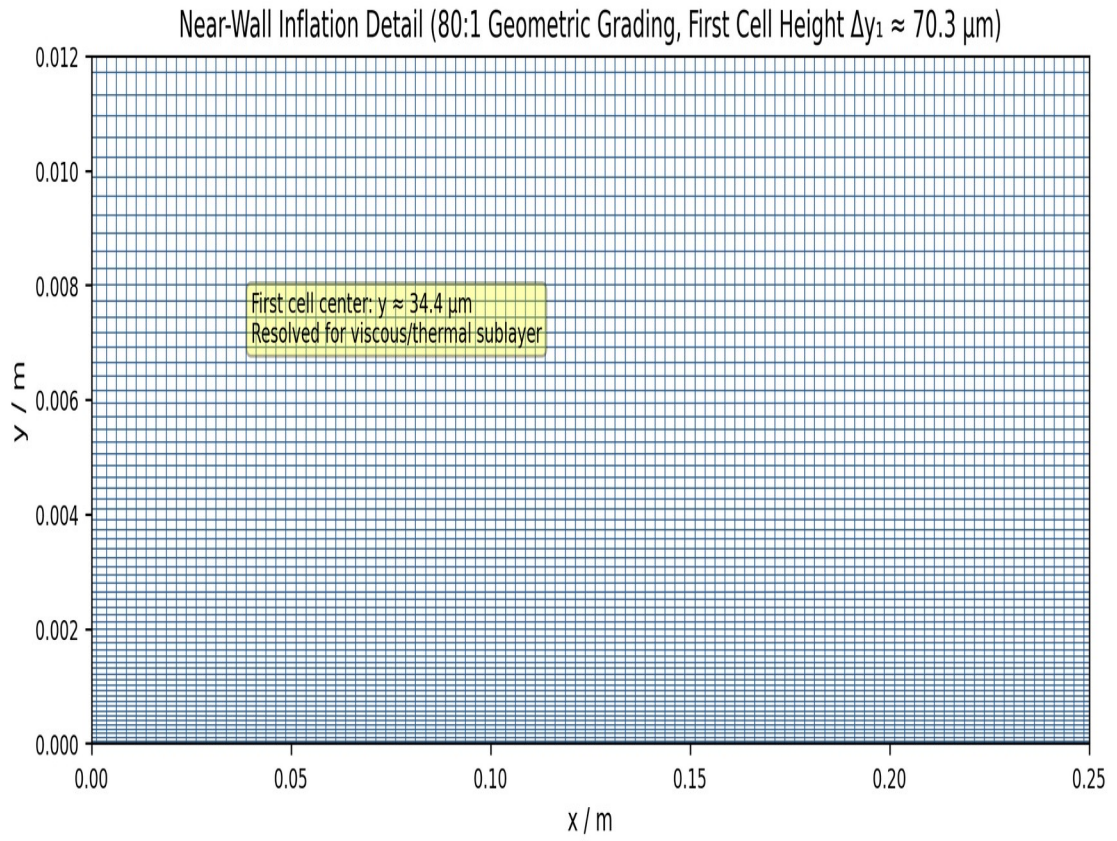
Fig. 1 Caption: what to look at, in one line.

Radiation sheds heat where convection weakens

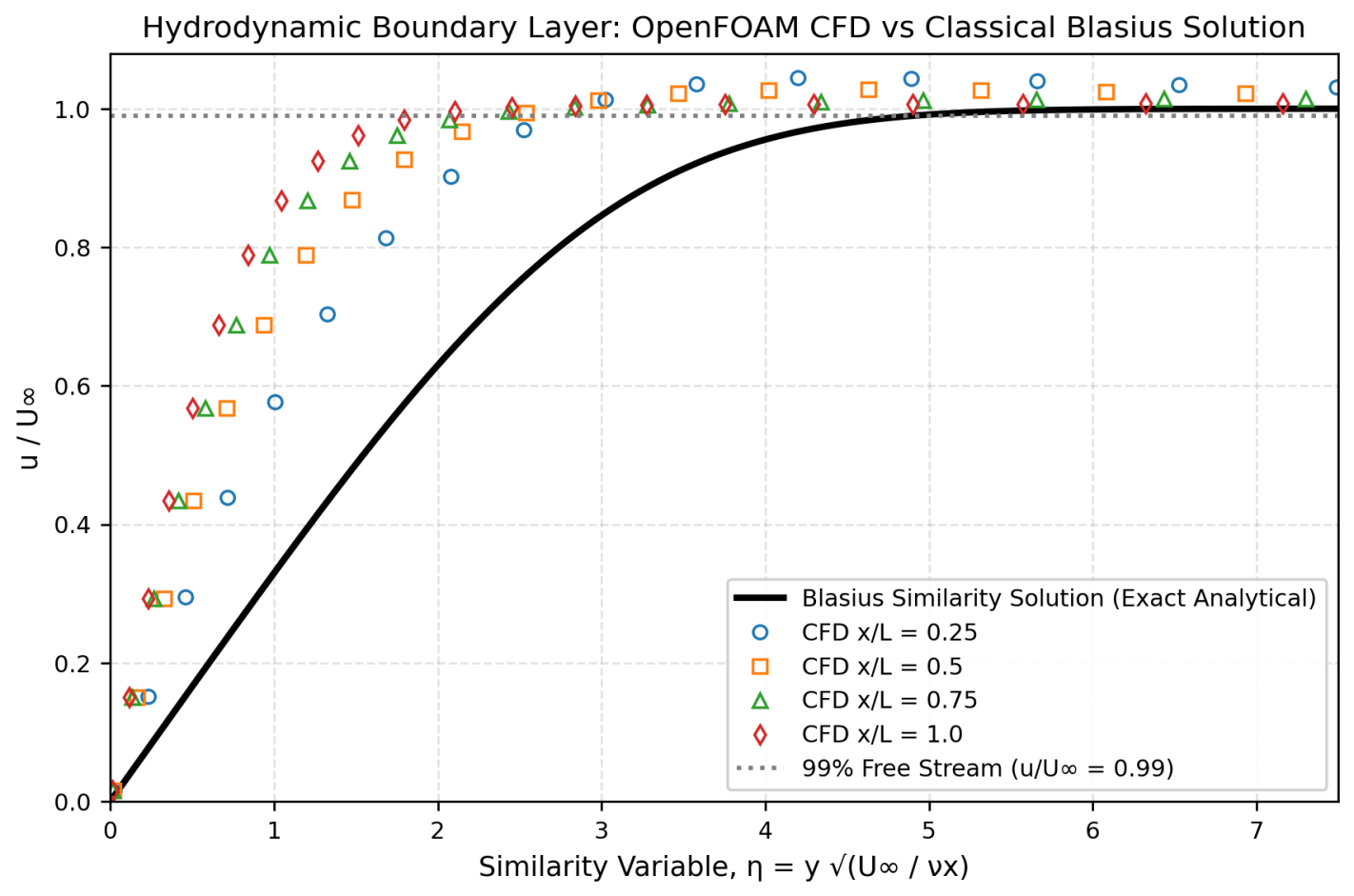
- Claim one, max 12 words
- Claim two, max 12 words
- Claim three, max 12 words

Setup table with mesh figure

Parameter	Value
Plate length	1.0 m
Free stream	2 m/s, 300 K
Heater flux	1000 W/m²
Emissivity	0 / 0.8
Grids	200×80, 400×160



Validation slide on dark background



0.11%

skin-friction error vs Blasius

[x%]

thermal profile error (report it)

[x]

grids used (3 needed for GCI)

Before and after comparison

1 Without radiation (ε = 0)

All 1000 W/m² leaves by convection

[metric]

2 With radiation (ε = 0.8)

Convection plus radiation share the load

[metric]

Limits and next steps, stated up front

Known limits

- Two-grid study only
- Constant properties
- Air treated as transparent

Next steps

- Third mesh, GCI error bars
- Global energy balance
- Variable-property run

[One-sentence conclusion and the ask]

[What you want from the audience: funding, review, collaboration]

github.com/Jersam-l/flat-plate-with-radiation-cfd | [email]